

Curriculum Format

Programme Name:

M. Tech. Computer Science & Engineering (In collaboration with Virusa (5 Year Integrated Programme))

Programme School:
Curriculum applicable to

Computer Science and Engineering (SCOPE)

(for admitted year xxxx-xx format):

AY 2023-24 onwards

Curriculum Version:

1.1

Approved by & Date

13th Academic Council, 28-12-2024

University Core Min. Credit Requirements ----->						102			
Course Code	Category	Course Title	Ver.	T	P	C	Pre/Co-Requisite	Anti-requisite	Course Discipline
	Projects and Internships	All are compulsory				28			
MIC5002		Internship/Senior Design project	1.0			20	None	None	Engineering
CAP5001		Capstone	1.0			6	None	None	Engineering
SPJ2001		Summer Intership/ Industry Certificate	1.0	-	-	2	None	None	Engineering
	Engineering Clinics Basket	All are compulsory				4			
ECS1001		Engineering Clinics - Embedded C & Arduino	2.0	0	4	2	None	None	Engineering
ECS1002		Engineering Clinics - Python & Raspberry Pi	1.0	0	4	2	None	None	Engineering
	Engineering Basket	All are compulsory				16			
SWE2005		Concepts of Object Oriented programming	4.0	3	2	4	None	None	Engineering
SWE1004		Introduction to Programming in Python	1.0	3	2	4	None	CSE1012	Engineering
SWE2001		Data Structures and its Applications	1.3	3	2	4	SWE1004/SWE2005	CSE2001	Engineering
ECE1005		Basic Electrical and Electronics Engineering	2.0	3	2	4	None	ECE1001, ECE1002	Engineering
	English Basket	Minimum of 6 Credits from courses in the basket				6			
ENG1005		Basic Communication in English	3.0	2	2	3	EPT (Not Cleared)	ENG1001	Humanities
ENG1006		Functional Communication in English	3.0	2	2	3	EPT (Cleared) /ENG1005	ENG1002	Humanities
ENG2004		Advanced Professional Communication	3.0	2	2	3	ENG1006	ENG2001	Humanities
	Humanities Basket	Minimum of 3 Credits from courses in the basket				3			
LIB1001		Anthropology - An Introduction	2.1	3	0	3	Anthropology of Technology	LIB1004	Humanities
LIB1002		Critical Thinking Skills	1.0	3	0	3	None	None	Humanities
LIB1003		Psychology - An Introduction	1.0	3	0	3	None	None	Humanities

Course Code	Category	Course Title	Ver.	T	P	C	Pre/Co-Requisite	Anti-requisite	Course Discipline
LIB1004		Sociology - An Introduction	2.1	3	0	3	None	LIB1001	Humanities
FRL1001		Basic French	2.0	3	0	3	None	None	Humanities
FRL1004		Basic Spanish	1.1	3	0	3	None	None	Humanities
FRL2001		Proficiency in French	2.0	3	0	3	FRL1001	None	Humanities
	Soft skills Basket	All are Compulsory				18			
STS0101		Introduction to Soft Skills	2.0	3	0	3	None	None	Engineering
STS1001		Introduction to Business Communication	1.1	3	0	3	STS0101	None	Engineering
STS1002		Fundamentals of Aptitude	3.0	3	0	3	STS1001	STS1004	Engineering
STS1007		Arithmetic problem solving	3.0	3	0	3	STS1002/STS1004	STS1009	Engineering
STS2006		Getting started to skill enhancement	2.0	3	0	3	STS1007/STS1009	STS2008	Engineering
STS2007		Enhancing problem solving skills	3.0	3	0	3	STS2006/STS2008	STS2009	Engineering
	Management Basket	Minimum of 5 Credits from courses in the basket				5			
MGT1001		Ethics and Values [Compulsory]	2.0	P/F		2	None	None	Management
MGT1002		Lean Start-up Management	1.1	3	0	3	None	None	Management
MGT1003		Fundamentals of Marketing	2.0	3	0	3	None	None	Management
MGT1004		Introduction to Organizational Behaviour	1.0	3	0	3	None	MGT1027	Management
MGT1005		Fundamentals of Creativity and Innovation Management	1.1	3	0	3	None	None	Management
MGT1006		Fundamentals of Operations Management	1.1	3	0	3	None	None	Management
MGT1007		Economics for Engineers	1.0	3	0	3	None	None	Management
MGT1008		Accounting and Finance for Engineers	1.1	3	0	3	None	None	Management
MGT1010		Fundamentals of Behavioral Finance	1.0	3	0	3	None	None	Management
MGT1015		Fundamentals of Consumer Behaviour	3.0	3	0	3	None	None	Management
MGT1020		Fundamentals of Services Marketing	2.0	3	0	3	None	None	Management
MGT1021		Fundamentals of Supply Chain Analytics	1.0	3	0	3	CSE1006	None	Management
MGT1022		Fundamentals of Insurance and Risk Management	1.0	3	0	3	None	None	Management
MGT1026		Basics of Investment Analysis and Portfolio Management	2.0	3	0	3	None	None	Management
MGT1027		Fundamentals of Team Building and Leadership	2.0	3	0	3	None	MGT1004	Management
MGT1028		Social Process and Social Change	1.0	3	0	3	None	None	Management
MGT1029		Basic Law for Engineers	2.0	3	0	3	None	None	Management
MGT1030		Fundamentals of Sustainability Management	1.0	3	0	3	None	None	Management
	Science Basket	All are compulsory				20			
MAT1008		Fundamentals of Calculus	1.0	3	2	4	None	MAT1001	Science
MAT1009		Applied Linear Algebra	3.1	4	0	4	None	MAT2005, MAT2010, MAT1106	Science
MAT1014		Fundamental Statistics	2.1	3	2	4	None	MAT1011	Science
PHY1005		Fundamentals of Engineering Physics	2.0	3	2	4	None	PHY1001, PHY1008, PHY1010	Science
CHY1008		Basic Chemistry and Environmental Studies	1.0	3	2	4	None	CHY1009	Science

Course Code	Category	Course Title	Ver.	T	P	C	Pre/Co-Requisite	Anti-requisite	Course Discipline
		Co/Extra Curricular [Compulsory]	2.0			2	None	None	
Programme Core Min. Credit Requirements ----->							48		
Course Code	Category	Course Title	Ver.	T	P	C	Pre/Co-Requisite	Anti-requisite	Course Discipline
SWE1002		Principles of Software Engineering	2.0	3	2	4	None	CSE1005	Engineering
SWE2006		Database Systems	1.0	3	2	4	CSE2002/SWE2001	CSE2007	Engineering
SWE2007		Introduction to Operating Systems	2.0	3	2	4	None	CSE2008	Engineering
SWE2009		Analysis of Algorithms	1.0	3	2	4	SWE2001	CSE3004, CSE3023	Engineering
SWE3001		Introduction to Computer Networks	1.1	3	2	4	SWE2005	ECE3003, CSE3003	Engineering
SWE3004		Front End Design and Testing	1.0	3	2	4	None	None	Engineering
SWE4006		Advanced Server Side Programming	1.0	3	2	4	SWE3004	None	Engineering
SWE4007		Software Application Architecture	1.0	3	2	4	SWE4006	None	Engineering
SWE4008		Application Development and Deployment Architecture	1.0	3	2	4	SWE4006	None	Engineering
ECE1006		Digital Logic and Microprocessors	1.1	3	2	4	ECE1005	ECE1003, ECE2004	Engineering
ECE2015		Computer Architecture	1.0	4	0	4	None	None	Engineering
MAT1007		Discrete Mathematics	2.0	4	0	4	None	MAT1003	Science
Programme Electives Min. Credit Requirements ----->							28		
Course Code	Category	Course Title	Ver.	T	P	C	Pre/Co-Requisite	Anti-requisite	Course Discipline
		Any course to fulfill the basket requirements							
CSE1006		Foundations for Data Analytics	1.1	2	2	3	None	None	Engineering
CSE1021		Foundations of Blockchain Technology	1.0	3	0	3	None	None	Engineering
CSE2009		Soft Computing	1.1	3	2	4	None	None	Engineering
CSE2011		Cyber Security and Digital Forensics	1.0	3	2	4	CSE1007/SWE3003	None	Engineering
CSE3001		Agile Development Process	1.1	3	2	4	CSE1005/SWE1002	None	Engineering
CSE3002		Artificial Intelligence	2.1	3	0	3	CSE1012/SWE1004	None	Engineering
CSE3008		Introduction to Machine Learning	3.0	3	2	4	None	None	Engineering
CSE3009		NoSQL Database	2.0	3	2	4	CSE2007	None	Engineering
CSE3012		Network Security	4.0	3	2	4	CSE3003/ ECE2006	None	Engineering
CSE3013		Secure Group Communications	1.1	3	0	3	CSE1007/SWE3003	None	Engineering
CSE3018		Software Configuration Management	1.1	3	0	3	CSE1005/SWE1002	None	Engineering
CSE3019		Software Quality and Reliability	1.0	3	0	3	CSE1005/SWE1002	None	Engineering
CSE3020		Product Definition and Validation	1.0	3	0	3	CSE1005/SWE1002	None	Engineering
CSE4003		Parallel Computing	2.1	3	2	4	CSE2008/ECE2002	None	Engineering
CSE4005		Data Warehousing and Data Mining	1.1	3	2	4	CSE1006	None	Engineering
CSE4006		Deep Learning	4.1	3	2	4	None	None	Engineering
CSE4007		Digital Image Processing	1.0	3	2	4	None	None	Engineering
CSE4008		Social Network Analysis	1.0	3	2	4	CSE1006	None	Engineering

Course Code	Category	Course Title	Ver.	T	P	C	Pre/Co-Requisite	Anti-requisite	Course Discipline
CSE4009		Wireless and Mobile Computing	1.0	3	0	3	CSE3003/SWE3001	None	Engineering
CSE4011		Internet of Things	1.0	3	2	4	None	None	Engineering
CSE4012		UI/UX User Experience Design	1.0	3	2	4	None	None	Engineering
CSE4013		Reverse Engineering	1.0	3	0	3	CSE1005/SWE1002	None	Engineering
CSE4014		Team Software Process	1.0	3	0	3	CSE1005/SWE1002	None	Engineering
CSE4015		Network and System Administration	1.1	3	2	4	SWE3001/CSE3003	None	Engineering
CSE4016		DataBase Administration	1.1	3	2	4	CSE2007/SWE2006	None	Engineering
CSE4017		Healthcare Analytics	1.1	3	0	3	CSE1006	None	Engineering
CSE4046		Design Patterns	1.1	3	2	4	CSE2005/SWE2005	None	Engineering
SWE2002		Human Computer Interaction	1.2	3	0	3	SWE1002	None	Engineering
SWE2003		Requirements Engineering Management	1.1	3	2	4	SWE1002	None	Engineering
SWE3003		Information and System Security	2.1	3	2	4	None	CSE1007	Engineering
SWE4003		Concepts of Mobile Application Development	2.0	3	2	4	SWE2005	CSE4002	Engineering
SWE4004		Cloud Computing and its Applications	1.1	3	2	4	SWE3001, SWE2007	CSE4001	Engineering
SWE4005		Storage Technologies and Management	1.0	3	0	3	SWE2006	None	Engineering
SWE4010		Formal Language and Automata Theory	1.0	4	0	4	None	CSE1008,CSE1012	Engineering
University Electives Min. Credit Requirements ----->						22			
STS3004		Introduction to Programming Skills [Compulsory for PAT]	1.1	3	0	3	STS2007	None	Engineering
STS4003		Enhancing Programming Ability [Compulsory for PAT]	1.0	3	0	3	STS3004/STS3005	None	Engineering
Any Course after fulfilling Programme Core & University Core requirements and without duplicity can be taken as University Elective									

Credits Summery	
University Core	Credits
Projects and Internships	28
Engineering Clinics Basket	4
Engineering Basket	16
English Basket	6
Humanities Basket	3
Soft skills Basket	18
Management Basket	5
Science Basket	20
Co/Extra Curricular	2
Total UC	102
Program Core	48
Program Electives	28
University Elective	22
Total Credits	200